Three-Dimensional Analysis of Hand and Finger Movements during Piano Playing

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The movements required for piano playing usually involve low impact loads that do not exceed physiologic limits of human body, but their repetition may provoke microtrauma leading to overuse injuries. Experience may allow a pianist to modify the motor patterns used for a performance, allowing the highest accuracy with minimum effort. In the present study, hand and finger movement patterns were analyzed in 19 pianists (8 concert players, 11 students and teachers) while they played 16 measures of a minuet. The three-dimensional coordinates of their right hand and fingers were obtained by a motion analyzer. Three-dimensional finger velocity was determined, unitary kinetic energy was computed, and movements were divided into useful (for sound production) and erratic (extraneous movements not used for sound production). The number of key presses for each pianist was counted, and single-finger unitary kinetic energy computed. On average, the concert players used more total unitary kinetic energy than the students and teachers ($p < 0.05$, Wilcoxon test), while the useful unitary kinetic energy was similar. The number of key presses for each finger did not differ ($p > 0.05$, chi-squared test). The useful unitary kinetic energy per single key press differed between groups ($p = 0.035$, with concert players greater than students and teachers, analysis of variance) and among the five fingers ($p = 0.008$, with second and first fingers larger). In conclusion, the same piano exercise was performed with different movement patterns depending on the pianist's experience. The patterns of extraneous hand and finger movements during playing could be investigated to assess their relationship to overuse injuries. *Med Probl Perform Art* 2007;22:18–23.

MUSIC PERFORMANCE is a human ability shaped by both individualistic, personal aspects and codified, normative aspects.[1] Cognitive and motor skills interact to provide a very particular form of communication,[2,3] where the same musical idea, codified by the composer, can be modeled and interpreted by the performer in numerous different performances.[1]

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The motor skills required for a successful musical performance combine natural body movements in complex patterns (or motor programs[1]) that are repeatedly reiterated within spatial and temporal constraints.[2,4] These movements usually require low impact loads that do not exceed the physiologic limits of human body in themselves (i.e., when they are taken one by one), but their continuous repetition may provoke microtrauma leading to overuse injuries.[5–7] The effect of these repetitive, low-impact stresses is well known to be at the origin of several work injuries,[8] and these disorders also could be found in professional musicians.[5,8] A detailed study of body movements during musical performance seems necessary to assess the biological and mechanical demands of this human activity.

In previous investigations, body movements during violin, guitar, and piano playing were studied in detail,[1,3–5,9–11] but only in some instances were the players asked to perform in conditions that actually mimicked their musical performance during daily training.[5,6] Also, data collection often focused on limited parts of the performance, without considering the global movements. In particular, piano playing appears to have been studied only within very restricted experimental conditions.

Additionally, in musical performance, as well as in all highly specialized motor activities in humans, increasing years of experience should enable the expert player to perform reproducible motion patterns[1,5,10] so that the highest accuracy can be obtained with the minimum effort.[3,10] Experience, therefore, should always be taken into consideration when assessing body movements, even if this seems to depend on the kind of stimulus. Although differences in performance levels across pianists depend on experience and practice for complex musical sequences,[1] in one study when the pianists were asked to perform a simple and basic exercise (such as a C-major scale), no strict relationship with the years of musical training was found.[11] Reproducibility appeared to be more related to the kind of musical performance currently played, with teachers and learners on one side and concert pianists on the other side.[11] Also, no correlation between performance and experience was found by Baader et al.,[2] who analyzed hand (bow) and finger movements in a group of violinists playing a simple sequence of ascending and descending tones.

In an earlier investigation by Sforza et al.,[11] technical problems required that the pianists play the two-octave scales with

a single hand, the same fingering was used, and the performance was paced by a metronome. All these constraints may have altered the actual performance and introduced extraneous factors that obscured the differences across performers; in particular, bimanual playing of scales is part of the standard training of pianists.[12] Additionally, only the useful (used for sound production) finger movements were quantitatively assessed.[11] Indeed, only global movements (including the extraneous finger movements, not useful for sound production) during playing could supply quantitative information about the load imposed on the fingers, hand, and wrist during playing.[5,6] In the past, extraneous, non-useful finger movements were used to assess their usefulness during learning,[10] to quantify the interdependence among fingers,[2,13] as well as to evaluate anticipatory movements.[2,4]

In the present study, three-dimensional hand and finger movement patterns were analyzed while pianists of different experience and professional background[11] played 16 measures taken from a minuet by Bach. The movements were recorded by an optoelectronic motion analyzer with passive markers, and no constraints on the actual hand and finger movements were imposed. The use of a complex musical sequence might allow us to estimate the actual load imposed on body structures during playing, as well as to assess the influence of experience on the actual movements. In particular, two groups of performers with different professional backgrounds were compared. Our hypothesis was that different patterns of movement would be used depending on the player's experience.

MATERIALS AND METHODS

Subjects and Experimental Task

Nineteen pianists (5 females, 14 males), aged from 8 to 69 years, volunteered for the study (Table 1). All were professionals with 2 to 62 years of experience. After the nature and possible risks of the study were described, written informed consent was obtained from each participant and from the parents of the participant younger than 18 years. The protocol used in the current study was approved by the local ethics committee, and it did not involve dangerous or painful activities.

According to their experience and personal description, the pianists were classified as students (3 female and 5 male, all full-time conservatory students), teachers (1 female and 2 male, full-time conservatory teachers), and concert players (1 female and 7 male, all graduates from an Italian conservatory who devoted themselves to concert playing).

The relative joint positions of the right hand and fingers were measured using a three-dimensional motion analysis system. On the right hand of each pianist, seven spherical, passive retro-reflective markers (diameter: 4 mm for the fingers, 6 mm for the hand) were positioned on the dorsum of a) the distal phalanx (on the nail) of the fingers, b) the middle of the third metacarpal, and c) the styloid process of the radius, using a bi-adhesive plaster. Markers' positions were carefully controlled to avoid any interference with playing.

TABLE 1. Characteristic of Pianists Participating in the Study

Pianists	Sex	Age (yrs)	Yrs of Piano Training
Students/Teachers			
S1	M	13	5
S2	F	8	2
S3	M	26	19
S4	F	21	8
S5	M	17	9
S6	F	9	3
S7	M	26	8
S8	M	22	7
T1	M	59	49
T2	F	26	20
T3	M	41	37
Mean (SD)		24.4 (14.8)	15.2 (15.1)
Concert Players			
C1	M	68	61
C2	M	36	28
C3	M	69	62
C4	M	31	24
C5	M	41	31
C6	F	34	25
C7	M	39	34
C8	M	40	32
Mean (SD)		44.8 (15.0)	37.1 (15.4)
Group Mean (SD)		32.8 (17.8)	24.4 (18.5)

The pianists were seated on a piano bench that was adjustable in height. In the same session and after a warming-up period, each pianist played 16 measures of a minuet by Bach on a Bechstein grand pianoforte. The 16 measures represent the first part of the minuet (32 measures in total). All the pianists knew the piece in advance, and they all could master the execution. The minuet was played with free fingering and free velocity; three repetitions were played by each pianist using the same fingering. The minuet spanned two octaves (14 notes).

Data Acquisition

An optoelectronic computerized instrument (Smart System, E-motion, Padova, Italy) was used for data acquisition. It allowed the automatic analysis of movement from the three-dimensional coordinates of the different hand landmarks as detected by six infrared-sensitive, charge-coupled device cameras.[11,14] The system had a sampling rate of 60 Hz, and it could follow fast and complex movements developing in the three spatial planes. The six cameras were positioned at various angles around the piano, four on the sides and two over the piano, covering a working area of 44 × 22 × 66 cm, to film the right hand of each pianist from different points of view (Figure 1).

Before each acquisition session, a metric calibration and correction of optical and electronic distortions were performed, with a resulting precision of 0.018%. During

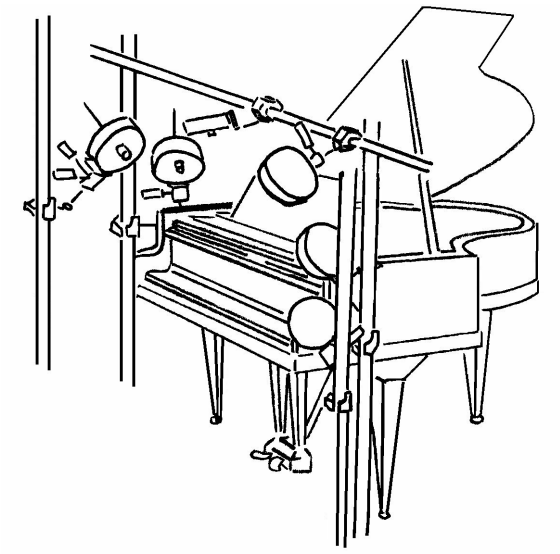

FIGURE 1. Experimental setting.

piano playing, special software recognized the coordinates of the center of gravity of each marker for any camera. Subsequently, all the coordinates were converted to real metric data, and a set of *x, y, z* coordinates for each landmark in each frame that constituted the movement was obtained.[11,14] All pianists began each minuet with a motionless hand, and this position was used as a reference for subsequent calculations.

Data Analysis

From marker coordinates, the movements during time were computed, and three-dimensional hand and fingers velocity obtained. The value was squared, and a "unitary kinetic energy" was obtained. Indeed, squared velocity (units: $m^2\ s^{-2}$) is equivalent to a unitary kinetic energy: energy is mass by squared velocity (units: $kg^1\ m^2\ s^{-2}$), which is divided by unitary mass (unitary energy, units: $kg^1\ m^2\ s^{-2}\ kg^{-1}$). This allowed: 1) the avoidance of negative values; and 2) the use of some measurement more similar to the actual muscular energy expenditure while playing.[10] A total unitary kinetic energy for each minuet was calculated and divided into those of useful movement (useful for sound production, displacements for and during key press, comprising the vertical displacements for actual key press and the useful finger movements like thumb-crossing) and erratic motions (extraneous movements not used for sound production).

The percent contribution of useful and erratic unitary kinetic energies was assessed. Inter-individual variability in unitary kinetic energy was assessed by calculating the coefficients of variation (percentage ratio of standard deviation to mean) among the three repetitions performed for each pianist.

Using the fingering of each pianist, the number of key presses for each finger was counted, and the percentage contribution (over a total of 56 notes) of each finger was assessed. Subsequently, single finger unitary kinetic energies (erratic and useful) were computed, as well as unitary kinetic energies for single key presses (from the percentage use of each finger). The number of teachers was too small for statistical comparisons, and the pianists were divided into only two groups: concert players and students/teachers (Table 1). Data were compared with analysis of variance, Wilcoxon rank test, and chi-squared test. Significance was set at 5% ($p < 0.05$) for all statistical comparisons.

RESULTS

On average, concert players were significantly older and had significantly more years of training than the students and teachers (Table 1, Wilcoxon rank test, $p < 0.05$).

The total unitary kinetic energy recorded for all hand and finger movements during the execution of the three repetitions of the 16-measure minuet was significantly larger in the concert players than in the students and teachers (Table 2, Wilcoxon rank test, $p < 0.05$). The difference was due to a significantly larger erratic unitary kinetic energy in concert players than in students and teachers ($p < 0.05$). In contrast, the useful unitary kinetic energy (that used for sound production) was similar in the two groups ($p > 0.05$). The percentages of useful and erratic unitary kinetic energy did not differ between the two groups. In the students and teachers group, no apparent relationship between age or body dimensions and performance was observed.

TABLE 2. "Unitary Kinetic Energy" Recorded for Hand and Finger Movements during Three Repetitions of the Minuet (16 Measures, 56 Notes) *

	Total Unitary Kinetic Energy ($kg^1\ m^2\ s^{-2}\ kg^{-1}$)	Erratic Unitary Kinetic Energy ($kg^1\ m^2\ s^{-2}\ kg^{-1}$)	CV (%)	Useful Unitary Kinetic Energy ($kg^1\ m^2\ s^{-2}\ kg^{-1}$)	Useful Unitary Kinetic Energy (%)	CV (%)
Concert players (n = 8)	190.25 (86.40)	172.50 (80.22)	5.72 (3.05)	17.75 (7.76)	9.70 (2.14)	9.25 (8.17)
Students and teachers (n = 11)	103.73 (39.78)	92.18 (34.46)	4.71 (3.66)	11.55 (6.93)	11.53 (4.40)	19.26 (14.02)
Comparison†	<0.05	<0.05	NS	NS	NS	NS

*Data expressed as mean (SD). "Erratic" movements are not useful for sound production; "useful" movements are performed for sound production. CV, coefficient of variation among the three repetitions.

†Comparison are expressed as p values, Wilcoxon rank test, with significance defined as $p > 0.05$; NS, not significant.

TABLE 3. Percent Use of Single Fingers while Playing 16 Measures of a Minuet (56 notes)

	Thumb	Index	Middle	Ring	Little
Concert players (n = 8)	20.8 (3.1)	25.5 (1.9)	26.6 (2.0)	16.1 (3.7)	11.2 (3.0)
Students and teachers (n = 11)	18.8 (3.4)	24.2 (3.3)	25.5 (2.4)	18.7 (3.1)	12.8 (3.6)

All values are percentages, expressed as mean (SD). No significant differences between groups (chi-squared test = 2.36, 4 *df*).

Intra-individual variability in the execution of the three repetitions of the minuet was somewhat larger in concert players than in students and teachers for the total unitary kinetic energy of erratic movements (approximately 1% more in concert players). A reverse trend was found for the percentage of useful unitary kinetic energy: on average, the concert players had coefficients of variation 6% smaller than those of the students and teachers. Nevertheless, the differences were not statistically significant ($p > 0.05$, Wilcoxon rank test).

The number of key presses for each finger was similar between the two groups of pianists: the index and middle fingers were the two most used fingers, whereas the little finger played only 11% (concert players) and 13% (students and teachers) of the 56 notes (Table 3, $p > 0.05$, chi-squared).

When the analysis was performed separately for each finger (Table 4), the total useful unitary kinetic energy differed significantly between the groups (two-way analysis of variance, factor group, $F = 5.76$, 1,85 degrees of freedom [*df*], $p = 0.019$, concert players larger than students and teachers), but not among the five fingers (factor finger, $F = 1.39$, 4,85 *df*, $p > 0.05$). The group × finger interaction was also not significant ($F = 1.1$, 4,85 *df*, $p > 0.05$). The useful unitary kinetic energy per single key press differed between groups (two-way analysis of variance, factor group, $F = 4.57$, 1,85 *df*, $p = 0.035$, concert players larger than students and teachers), as well as among the five fingers (factor finger, $F = 3.72$, 4,85 *df*, $p = 0.008$, second and first fingers larger), without group x finger interactions ($F = 1.18$, 4,85 *df*, $p > 0.05$).

DISCUSSION

Piano playing represents a highly skilled and sophisticated task that requires exact coordination between motor and sensorial skills.[4,15] In the players, the anatomical and biomechanical characteristics of the hand and finger joints and muscles have to be used according to performance constraints.[3]

In the current study, the right hand and finger movements of 19 professional pianists were assessed while they played 16 measures of a minuet without temporal constraints. The players were asked to play bimanually, so no external limitations were imposed on the execution. Additionally and in contrast with previous studies,[2-6,10,11] no fixed notation was forced and no fixed tempo was imposed on the players. Indeed, a personal internal rhythm influencing musical performance has already been hypothesized.[1,15]

The pianists were all classified as "professionals," even though eight of them (the students) were not conservatory graduates; however, they were all full-time conservatory students and devoted most of their time to piano playing. No physical constraints were imposed on the hand and finger movements, and the motion was detected by an optoelectronic instrument with passive markers. The markers were small (the largest were 6 mm in diameter) and practically weightless, allowing the unimpeded analysis of young people with small hands and fingers.[11,16] A similar motion analysis system had been used previously for simple hand and finger movements,[17–19] for keyboard typing,[20] as well as for piano and violin performance,[2,4-6,11] and negligible skin movement relative to the bone support was demonstrated.[18]

The biomechanical model of the hand and fingers was very simple, with a single marker for each finger considered as a rigid body. Indeed, most previous three-dimensional analyses of musical performance used one marker per finger only,[2,3,5,6,11] and only Engel et al.[4] positioned two markers per finger (distal interphalangeal joint and metacarpophalangeal joint). In that study, the motor task required of the pianists was simpler than the current one; they had to play only musical phrases excerpted from classical pieces, and the movement spanned seven notes only. Additionally, Engel et

TABLE 4. Useful "Unitary Kinetic Energy" Recorded for Each Finger during Three Repetitions of the Minuet

	Thumb	Index	Middle	Ring	Little
Concert players (n = 8)					
Total	4.473 (1.793)	4.248 (2.260)	2.938 (2.140)	2.579 (1.105)	3.351 (2.460)
Single key press	0.967 (0.502)	1.093 (0.627)	0.801 (0.606)	0.423 (0.189)	0.376 (0.312)
Students and teachers (n = 11)					
Total	2.939 (3.106)	2.625 (2.284)	1.741 (1.470)	3.333 (2.215)	1.790 (0.759)
Single key press	0.614 (0.719)	0.676 (0.674)	0.437 (0.377)	0.610 (0.374)	0.227 (0.113)

All values are $kg^1 m^2 s^{-2} kg^{-1}$, expressed as mean (SD).

al.[4] used a digital piano, which is smaller than the grand piano used in the present investigation; in the current study, the dimension of the piano limited the position of the cameras used for movement recording. More complex biomechanical models (up to 19 markers per hand) have been used for the analysis of single finger motions only, without musical performances.[19]

The current system allowed for the detection of both useful (movements made for sound production) and extraneous movements, and the interrelationship between these movements was the principal aim of the study. In a previous study, Parlitz et al.[10] applied a force measurement system to some piano keys and compared forces generated by expert and amateur pianists playing the same musical fragments at the same tempo and loudness. The authors found that the amateur players applied greater and longer forces to the keys than the expert players. Unfortunately, the authors did not assess actual movements while playing. Only Baader et al.[2] and Sforza et al.[11] compared players of differing experience during playing of a simple and basic musical exercise, but they did not find a strict relationship between movement performance and musical experience (years of musical training). A limitation of these two studies may have been the use of too simple stimuli for the most expert players.

The current study used a more complex piece, which may represent a good compromise between players of differing experience and training. In particular, Sforza et al.[11] found that teachers and musical students were more similar in the reproducibility of their performance than concert players. Their explanation was that a beginner, as well as a teacher, has to control each and all parts of the playing hand before concentrating on other aspects of their performance.[2,3] In contrast, a concert pianist is used to playing more complex sequences, and their focus could be on other aspects of musical performance.[1,3]

To investigate differences in performance further, and following the results of the previous investigation,[11] we divided the present performers into two groups: the concert players and the students and teachers. For the conservatory graduates, the classification was personal: without knowing the actual scope of the study, they were just asked which label best described their current professional situation.

On average, the concert players performed the minuet with different hand and finger movements than the students and teachers; their total unitary kinetic energy, as well as their unitary kinetic energy of erratic (extraneous) movements, was significantly larger for the hand and fingers together (Table 2). In contrast, no differences were found for the useful unitary kinetic energy (Table 2); the two groups used similar proportions of useful vs. erratic energies. Useful energy could be equated to the force actually measured in previous studies.[10] The present group of 19 pianists, therefore, seem to be more homogeneous than that analyzed by Parlitz et al.,[10] and the definition of "professionals" applied also to conservatory students could be correct, at least from this point of view. Useful unitary kinetic energy probably depends more on the codified aspect of the musical piece, and less on the personal interpretation.[1] The personal (and therefore the one linked to experience) aspect of performance might be reflected by the erratic movements of hand and fingers, which could be due to anticipatory movements,[2,4] to the possibility of expressing different forces according to the position of the nonperforming fingers,[13] as well as to strategies for minimizing stress.[5]

The two groups of pianists did not differ in the number of key presses for each finger (Table 3). Anatomical constraints and digit interdependence limit the possibility of motion,[2,3,19] and fingering, even personal, always follows general rules.[1]

Homogeneity also was found for intra-individual variability, as expressed by the coefficients of variation of useful and erratic unitary kinetic energies. Indeed, the concert players were somewhat (not significantly) more reproducible in their useful unitary kinetic energy as compared to the students and teachers, and this might point to either a better control of key press or better fluidity of execution.[1]

Differences between groups were also found for the useful unitary kinetic energies of single fingers (Table 4). Significant inter-finger differences were detected for the useful unitary kinetic energy per single key press; in both groups of pianists, the little finger was moved with approximately one third of the unitary kinetic energy of the thumb and index fingers. The lack of significant group × finger interactions points again to the anatomical and musical constraints that govern performance.[1-3,19] The only finger that appeared to move somehow differently between groups was the fourth (ring finger); this finger was used less, and it was moved with greater total unitary kinetic energy but with less useful unitary kinetic energy by the concert players than by the students and teachers. Both unitary kinetic energies were also more homogeneous (smaller coefficients of variation) within the concert players group than within the student/teacher group. The difference may be an effect of training.

During musical performance, the patterns of movement also could result in unnatural motions, with joint positions that are not usually employed during everyday life.[3,5,6] Additionally, the hands and fingertips of the players endure repeated dynamic loadings that are confined to a small area.[9,21] Considering that a professional pianist spends 6 hours a day on the keyboard, the repeated low-impact stress may lead, in some instances, to the same occupational disorders found in typists and laborers using hand-held vibratory tools.[5-8,10,21] Vascular, sensorineural, and musculoskeletal disorders have all been described as a consequence of this repeated loading, with pathologies such as carpal tunnel syndrome, hand-arm vibration syndrome (Raynaud's phenomenon), and flexor tenosynovities.[21,22] In particular, a recent occupational analysis found that the risk of a string player (violoncello) paralleled that of industrial ironing or marble polishing.[8]

Since the last decades of 19th century, several European musical schools developed didactic methods in which hand and finger movements at the keyboard were dictated by anatomical and physiologic considerations, in the attempt to abolish (or limit) occupational disorders for pianists.[23–25]

Indeed, didactic techniques should always teach the students how to reduce—or better, to avoid—professional disorders.[1,5] The use of a wrong technique may accelerate the development of overuse stress injuries; in contrast, correct playing technique could protect the player.[7] For instance, an efficient performance, with a limited energy expenditure, could be advantageous for professional pianists who spend several hours a day on the instrument.[10]

In conclusion, the same piano exercise was performed with different movement patterns that were in part related to the pianist's experience, thus confirming our initial hypothesis. During playing, pianists could use different techniques to reach the same results—e.g., leaving the nonperforming fingers in contact with the keyboard, as students and teachers did, or lifting them away from the keys, using more extraneous movements, as concert players did. The actual effect of these extraneous motions on the body should be assessed to verify their relationship to overuse injuries.[7] From a theoretical point of view, the use of more energy for body motion should implicate the occurrence of more stress, with a possible development of overuse injuries.[1,5–8,10] Nevertheless, patterns of extraneous finger movements during playing could be investigated in more depth; they might be a mirror of personal "interpretation" of sound production.

Future studies could be performed with a larger number of pianists, especially with more piano teachers and with students of different levels. Additionally, a wider set of musical pieces could be selected to allow a better identification of the characteristics of each group. If the current technical difficulties are to be overcome, then both (right and left) hands should be assessed,[2,3,5,6,12] and the fingers should also be identified by a larger number of markers, without unnecessary biomechanical approximations.[9,19] The resulting data could provide the basis for developing didactic techniques that reduce the possibility of overuse injuries in pianists.[5]

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